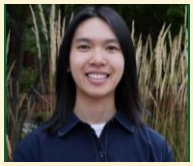
Evaluating Automated Methods to Bypass Invisible CAPTCHA

Anirudh Maramraj

Bishesh Joshi

David Nguyen



Diemmi Dao

Hiten Chintalapalli

Joseph-Vinh Nguyen

SMEs: Dr. Jair Feldens Ferrari

CHALLENGE

Today, it is challenging for internet-based services to ensure reliable and secure services to only legitimate users. The objective of this project is to develop an automated Python-based program that is able to successfully bypass invisible CAPTCHA without being detected as a bot. Furthermore, the aim of this project is to highlight the vulnerabilities in various CAPTCHA vendors and their current implementations across the web to reveal the current security status and the need to further secure these implementations.

Anirudh Maramraj

Fairfax, VA.

Aspiration: I aspire to engage in the legal aspects of cybersecurity. Though there are already numerous laws in place, cybersecurity is an ever-expanding field, and I believe there must be adequate policies that keep pace.

Class Comment: This class has been useful in giving me a chance to apply all the knowledge I have gathered over the past four years, and how well I can apply what I have learned in a team setting

Bishesh Joshi

Fairfax, VA.

Aspiration: I aspire to continue my education and career in cybersecurity, where I intent to protect and secure systems.

Class Comment: This class has been beneficial in teaching me the process of conducting research, applying it to a real-world scenario, leveraging previous coursework, and writing a research paper.

David Nguyen

Winchester, VA.

Aspiration: I aspire to continue to learn and expand my skill in secure programming and agentic AI/ML. I am hoping to contribute my skills to the United States Military.

Class Comment: This class has given me an opportunity to grow my programming skills, as well as enhancing my critical thinking skills.

Diemmi Dao

Fairfax, VA.

Aspiration: I aspire to work and gain more experience in artificial intelligence and digital forensics industries. I intend on continuously enhancing and learning new skills.

Class Comment: This class has been a valuable project-based experience with the chance to solve a real open-ended problem using skills that I have been honing to prepare for the work force.

Hiten Chintalapalli

Fairfax, VA.

Aspiration: I aspire to use the offensive and defensive security principles that I acquired here at George Mason University to pursue my career in Cybersecurity with a specialized focus in network security.

Class Comment: This class has given me insight into what it is like to work within a team and engage in a real-world project. I am looking forward to using this experience in my career.

Joseph-Vinh Nguyen

Fairfax, VA.

Aspiration: I aspire to build a cybersecurity career in the intelligence community, applying my skills acquired at GMU to national network defense and cyber operations.

Class Comment: This class has exposed me to the lifecycles of long-term, team-based projects that provide developments to real-world technologies.

Project Sponsor: Dr. Mingkui Wei, Cybersecurity Engineering Department at GMU

College of Engineering and Computing
CYBER SECURITY ENGINEERING
George Mason University

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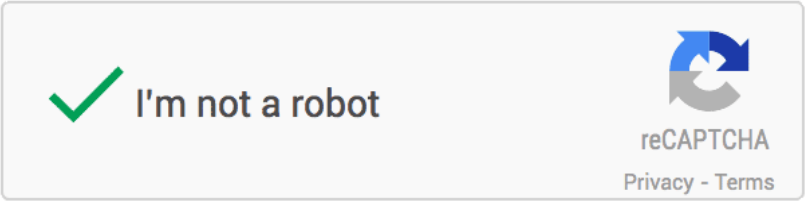
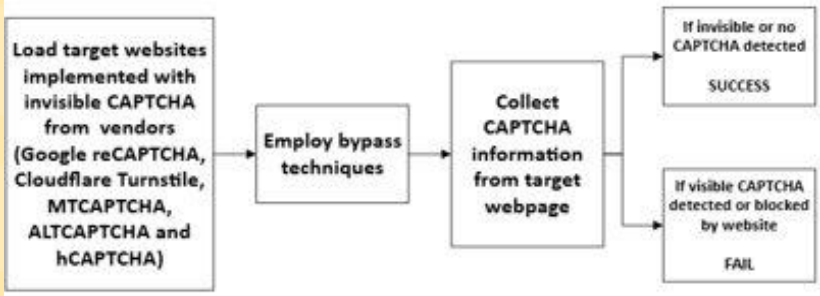


Currently, there are numerous ways to automatically bypass visible CAPTCHA formats such as typing in scribbled text, selecting boxes of a gridded image, or clicking the "I'm not a robot" checkbox. However, there is few research on how secure invisible or behavioral CAPTCHAs are against advanced automated attacks. Our project aims to explore automated attacks to see if invisible CAPTCHA implementations are currently keeping web services secure enough against automated bots.

Our deliverable is a Python-based program integrated with automated web scraping libraries Scrapy and Selenium and features third-party ScrapeOps Proxy API. Multiple websites integrated with invisible CAPTCHAs are used as a testing array to see if our bot is able to capture the web content without being blocked.

Changing various features such as user agent headers allows for the bot to be seen as a different device and depending on the website's configurations, may allow certain devices to explore the webpage freely.



Scanning the webpage for any active CAPTCHAs allows us to determine whether the website allowed or blocked the bot to view the content.

Picture source: https://developers.google.com/recaptcha/docs/versions, https://www.scrapy.org/

Project sponsor: CYSE, GMU